

2. **Group ID**: Captures the **Group ID** to which a groupcast message was sent.
3. **Source Node ID**: The **Source Node ID** enclosed by the sender of a groupcast message.
  - Together, **Fabric Index**, **Group ID** and **Source Node ID** comprise a unique identifier that upper layers may use to understand the source and destination of groupcast messages.
4. **Source IP Address**: The unicast source IP address for the originator of the message.
5. **Source Port**: The source port for the originator of the message.
  - The source IP address and port MAY be used for unicast responses to group communication peers, as are required for the **Message Counter Synchronization Protocol**.
6. **Operational Group Key**: The **Operational Group Key** that was used to encrypt the incoming group message.
7. **Group Session ID**: Records the **Group Session ID** derived from the **Operational Group Key** used to encrypt the message.

Once a Groupcast Session Context with **trust-first** policy is created to track authenticated messages from a given Source Node ID, that record SHALL NOT be deleted or recycled until the node reboots. This is to prevent replay attacks that first exhaust the memory allocated to group session counter tracking and then inject older messages as valid, and enforces that trust-first authentication works as intended within the full duration of a boot cycle. Any message from a source that cannot be tracked SHALL be dropped.

#### 4.16.2. Sending a group message

To prepare a multicast message to a Group ID with a given **GroupKeySetID** and IPv6 hop count parameter, the Node SHALL:

1. Obtain, for the given **GroupKeySetID**, the current **Operational Group Key** as the Encryption Key, and the associated **Group Session ID**.
  - a. If no key is found for the given **GroupKeySetID**, security processing SHALL fail and no further security processing SHALL be done on this message.
2. Perform **Section 4.7.1, “Message Transmission”** processing steps on the message with the following arguments:
  - a. The Destination Node Id argument SHALL be the **Group Node Id** corresponding to the given Group ID.
  - b. The Session Id argument SHALL be the Group Session ID from step 1.
  - c. The Security Flags SHALL have only the **P Flag** set.
  - d. The transport SHALL be a site-local routable IPv6 interface.

Next, prepare the message as an IPv6 packet:

1. Set the private, secured message from step 2 above as the IPv6 payload.
2. Set the IPv6 hop count to the value given.
3. Set the IPv6 destination to the **Section 2.5.6.2, “IPv6 Multicast Address”** based on the provided destination Group ID, Fabric ID, and **GroupKeyMulticastPolicy** of the group key.